Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Equilibrium

Meaning in my words How would you explain equilibrium to a friend?	**How can you tell?** A cart rolls straight at steady speed. What must the horizontal forces add up to?
Example or sketch Sketch the horizontal forces on a cart with a 3 N pull right and 3 N friction left.	**Non-example and why** A tossed ball is briefly still at the top. Is it in equilibrium there?

Balance

Meaning in my words How would you explain balance to a friend?	**How can you tell?** Two cables suspend a sign at rest. What must their upward components add up to?
Example or sketch A sign weighs 12 N and two cables share the upward support equally. What is each vertical contribution?	**Non-example and why** The horizontal components cancel. Is that enough to show equilibrium?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Symmetry

Meaning in my words How would you explain symmetry to a friend?	**How can you tell?** Two identical cables hold a centered sign at equal angles. What tells you their tensions match?
Example or sketch Sketch equal tension arrows. What happens to the two horizontal components?	**Non-example and why** One cable is much steeper. Can you still assume the tensions are equal from symmetry alone?

Rest

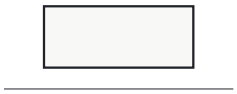
Meaning in my words What does it mean for an object to be at rest at an instant?	**How can you tell?** At the top of a ball's straight upward toss, what is zero and what is still acting?
Example or sketch How would you draw velocity and acceleration at that instant?	**Non-example and why** "It stopped, so the forces must balance." What does the next instant show?

Doodle the idea

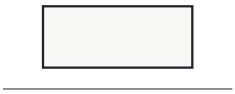
SECTION B / DRAW WITH YOUR TEACHER Name _____

Sketch two systems with balanced forces: one at rest and one moving steadily. Label the common acceleration.

At rest



Moving steadily



My marks mean

Stopped or coasting?

18

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Distinguish zero acceleration from zero velocity.

CHOOSE ONE JOB TOGETHER

FLIGHT CONTROLLER ☐

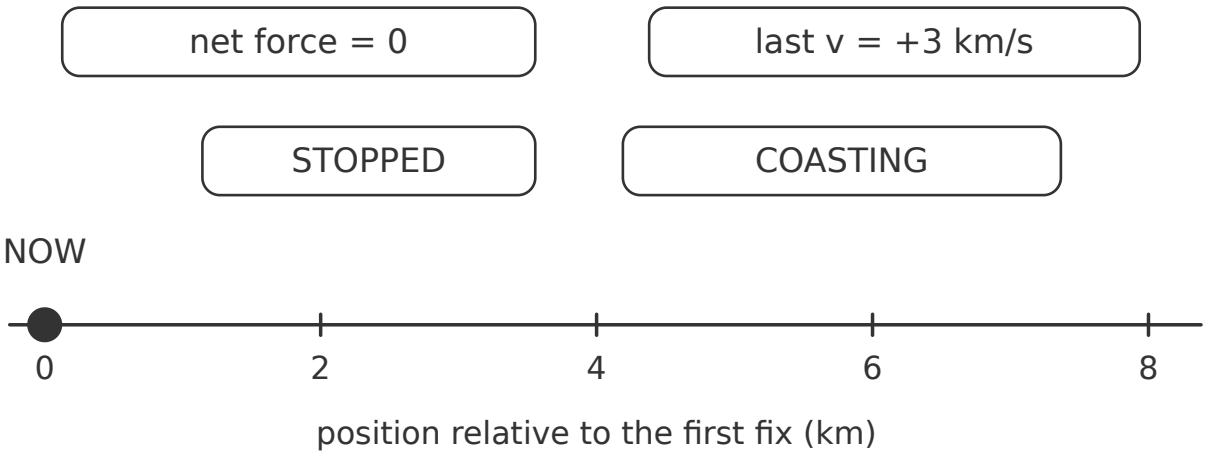
Choose the status graphic. Mark where XJ should be two seconds later relative to the first sighting.

BROADCAST PRODUCER ☐

Choose the on-air label. Explain why zero acceleration and zero speed do not mean the same thing.

Point to a clue. Keep your first marks; add the update using the color key.

The flight display reads "net force zero." XJ's last velocity was +**3 km/s**. Two graphics say STOPPED and COASTING.



01 OUR FIRST CALL

Which status should go live? Use a second position to support your call.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Can the old status describe the next interval? Explain what the new report changes.

Switch speakers. In two dimensions, what must the force sums be along both axes for the forces to balance?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks